Monocular Image based High-Altitude UAV Position Estimation Algorithm

Presenter : Kyeong Wook Seo

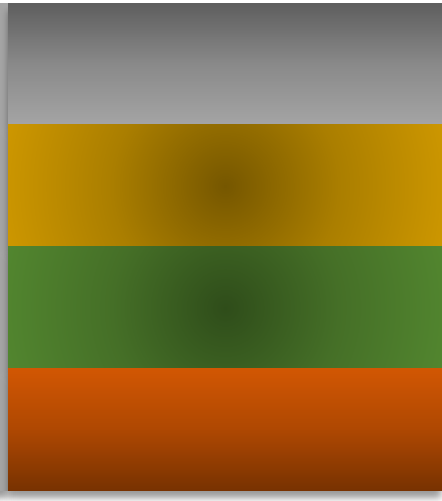
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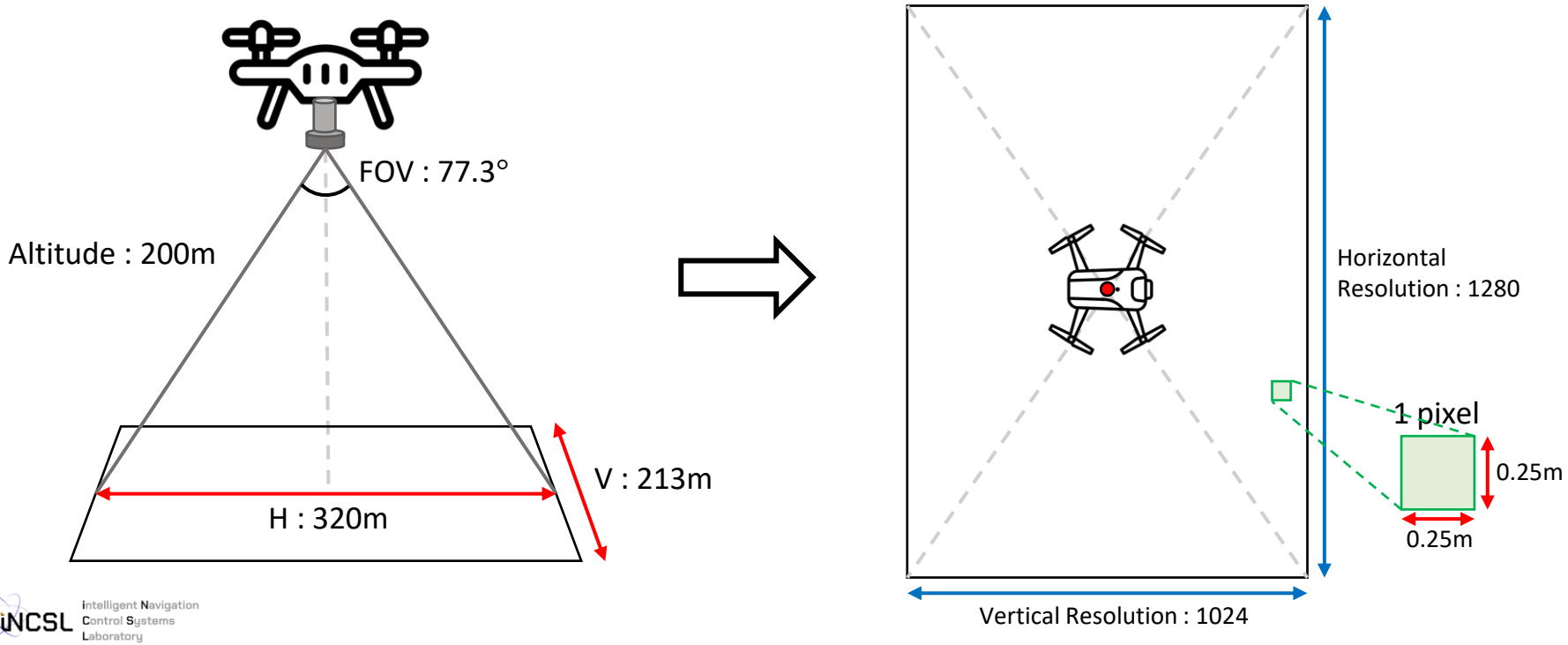
- Research Topic
- Vision based Navigation System
  - 2-D Image Processing
  - Multi State Constraint Kalman Filter
- Future Work
- Q & A

RESEARCH TOPIC



RESEARCH TOPIC (1/4)

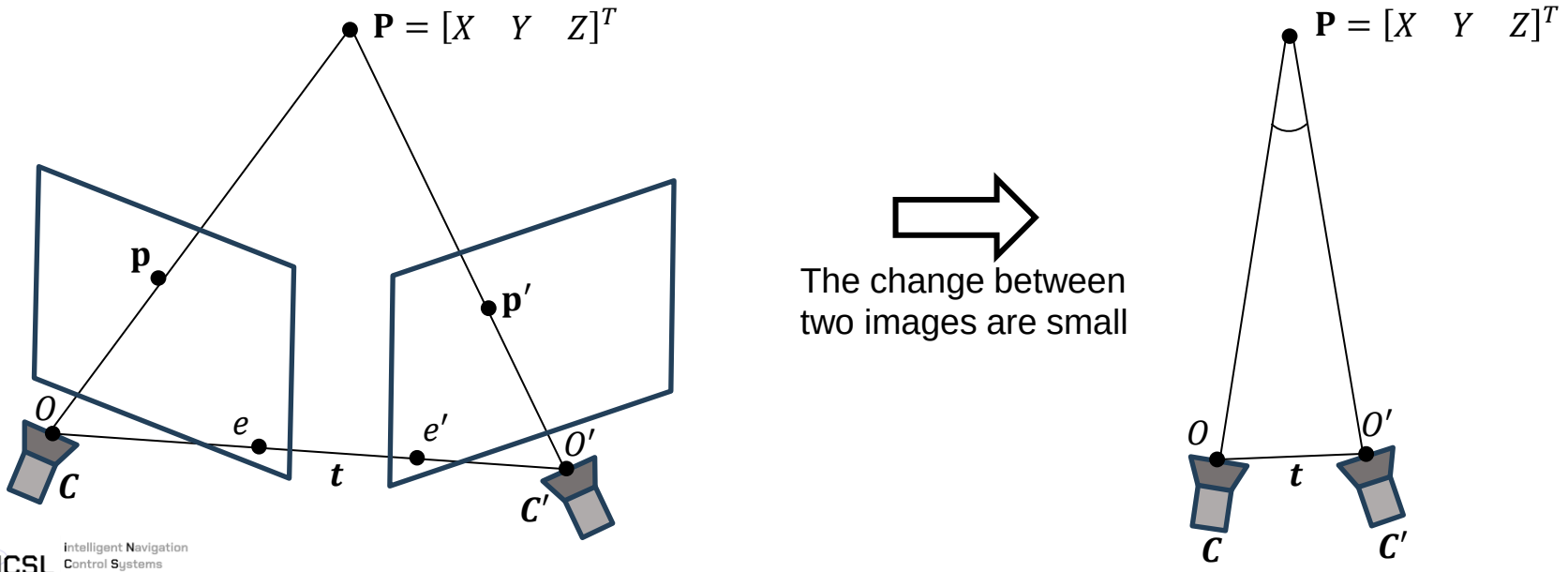
- In the case of UAVs(Unmanned Aerial Vehicles) operating at **high altitudes**, the drone's movement at relatively high altitudes may result in **minimal changes** in the observed images of the ground.
- Drone flying at an altitude of 200m would need to move approximately 0.25m for a 1 pixel change to be observed in the image.(altitude=30m → 1 pixel=0.0375m)



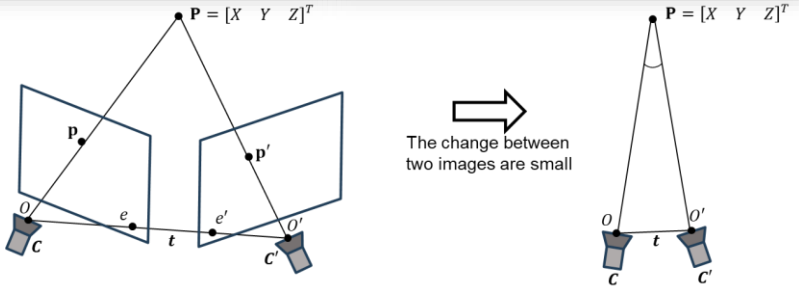
RESEARCH TOPIC (2/4)

$\mathbf{P}$	A point in 3D space
p, p'	The points projected onto the image
e, e'	Epipole
O, O'	The origin of the camera
$\mathbf{R}, \boldsymbol{t}$	Rotation Matrix & translation Vector
$\boldsymbol{C}, \boldsymbol{C}'$	Camera coordinates

- Epipolar Geometry
  - The geometric relationship of matched feature points in two images acquired from different viewpoints of the same feature point.
  - The points projected onto the images can be represented using the essential matrix and the fundamental matrix to express the transformation relationship.
  - In the figure below, $\boldsymbol{C}$ and $\boldsymbol{C}'$ represent the camera frames at specific times k and $k+1$, respectively



RESEARCH TOPIC (3/4)



- Essential Matrix
  - The matched points in images taken from any location can be expressed by the relationship shown in Eq. (1).

$$\mathbf{p'}^T\mathbf{E}\mathbf{p} = [u' \quad v' \quad 1]\mathbf{E}\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = 0 \qquad \cdots(1)$$

  - Point p' projected from the previous step can be represented from the 2-D position p using rotation matrix R and translation vector t as shown in Eq. (2).

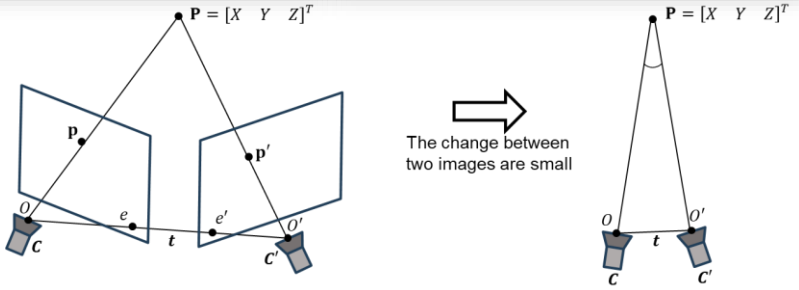
$$\mathbf{p'} = \mathbf{R}\mathbf{p} + \boldsymbol{t} \qquad \cdots(2)$$

  - When the difference between $\boldsymbol{C}$ and $\boldsymbol{C}$' is small, it is impossible to compute the essential matrix($\mathbf{p} \approx \mathbf{p'}$).
- Fundamental Matrix
  - This represents the geometric relationship between the actual pixel coordinates of two images, including camera intrinsics shown as Eq. (3).

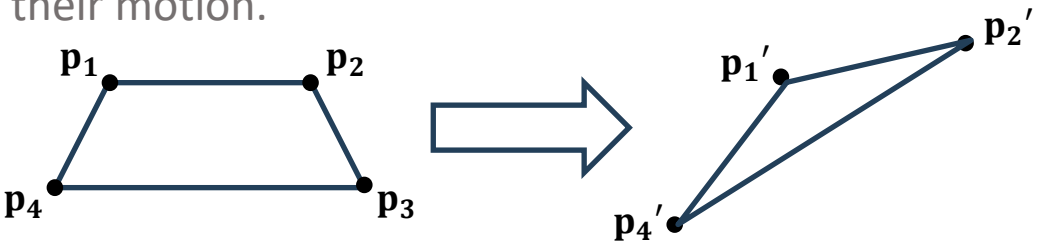
$$\mathbf{F} = \left(\mathbf{K}^{\boldsymbol{T}}\right)^{-\mathbf{1}}\mathbf{E}\mathbf{K}^{-\mathbf{1}} \qquad \cdots(3)$$

  - Since the essential matrix cannot be computed, similarly, the fundamental matrix cannot be obtained either.

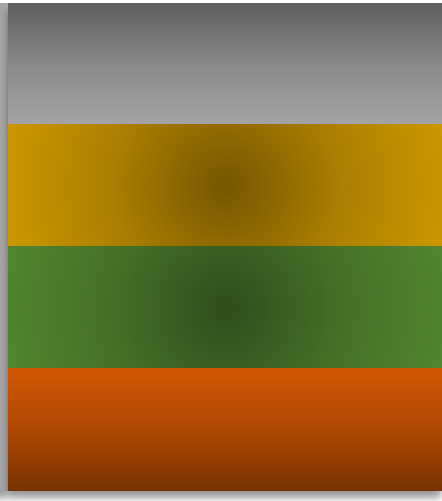
RESEARCH TOPIC (4/4)



- Additional Constraint
  - When attempting to compute the essential matrix for individual feature points, the mentioned issue arises.
  - Therefore, it is necessary to compute the essential matrix and fundamental matrix by connecting feature points or comparing specific terrain features.
- Compensation based on a dynamic model for the variations caused by vibrations
  - Setting a certain threshold based on the vibrations of roll and pitch measured through an accelerometer.
- In case of abrupt changes in orientation leading to the loss of previously detected feature
  - Designing MSCKF as a navigation filter to track feature points in case of significant changes in their motion.



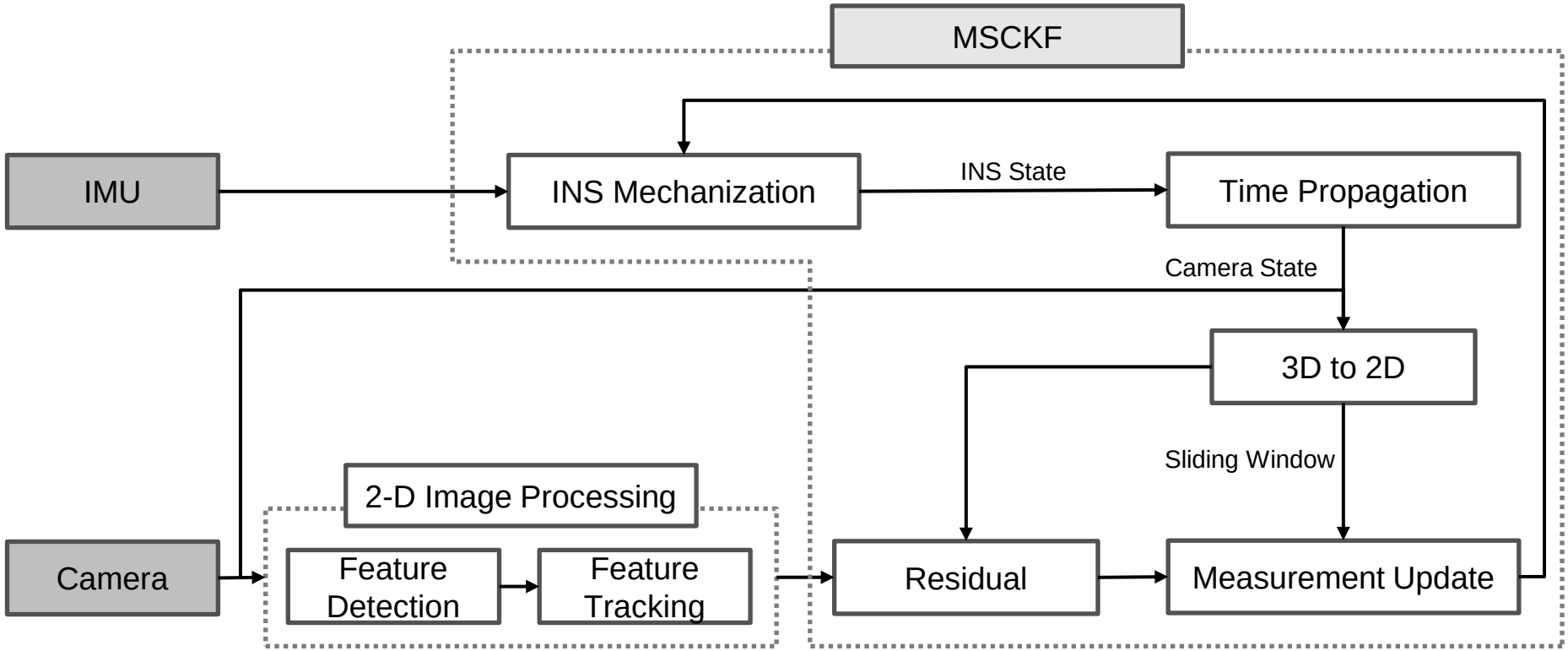
VISION BASED NAVIGATION SYSTEM



- Navigation System Overall
- 2-D Image Processing
- Multi State Constraint Kalman Filter

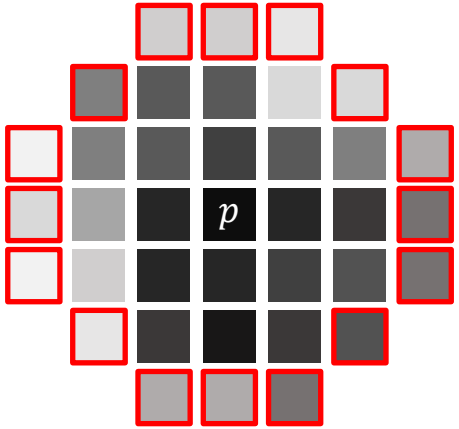
NAVIGATION SYSTEM OVERALL

- Localization system block diagram using a monocular camera with MSCKF



2-D Image Processing – Feature Detection

- An algorithm capable of real-time processing with a fast enough detection speed.
- Identifies feature points with a high matching score among the detected ones.
- FAST Feature Detection
  - Selecting an arbitrary pixel p in the image, and the brightness at this pixel is defined as I_p.
  - Setting a circle consisting of 16 pixels around pixel p as the reference.
  - If among the selected 16 pixels, there are n pixels with brightness values in the range of $I_p \pm t$ based pm the detection threshold t, it is considered a corner.
- Non-maximal Suppression
  - For all detected feature points, compute the score function $\boldsymbol{V}$ (where $\boldsymbol{V}$ is computed for the adjacent two feature points among the 16 surrounding p) and discard the smaller one.



2-D Image Processing – Feature Tracking

- Optical Flow
  - 이전 프레임과 현재 프레임 간의 명암 차를 이용하여 픽셀의 이동을 추출
  - Lucas Kanade method
    - 픽셀 p를 중심으로 하는 window 영역의 optical flow는 동일하다
    - 앞서 추출된 코너를 기반으로 추적하기 때문에 연산량이 적다.
    - 좁은 지역의 윈도우를 사용하기에 윈도우 크기보다 큰 움직임은 계산할 수 없다.
  - Horm-Schunk method
    - 영상 전체 픽셀의 움직임을 전역적으로 고려하는 방식
    - LK method에 비해 비교적 정확하다.
    - 모든 픽셀의 변화량을 계산하므로 다소 긴 연산 시간을 소모한다.

- Predict feature points
  - Homography Estimation
    - 두 이미지의 변환(rotation, translation) 관계를 통해 특징점의 위치 추정

- RANSAC → Delete Outlier

MULTI STATE CONSTRAINT KALMAN FILTER

- State vector
  - IMU state vector :

$$\hat{\mathbf{x}}_{IMU} = \begin{bmatrix} \hat{\mathbf{p}}^T & \hat{\mathbf{v}}^T & \hat{\boldsymbol{\psi}}^T & \hat{\mathbf{b}}_a^T & \hat{\mathbf{b}}_g^T \end{bmatrix}^T \in \mathbb{R}^{15\times 1} \quad \cdots(4)$$

  - MSCKF state vector :

$$\hat{\mathbf{x}}_k = \begin{bmatrix} \hat{\mathbf{x}}_{IMU}^T & \hat{\mathbf{p}}_{C_1}^T & \hat{\mathbf{v}}_{C_1}^T & \hat{\boldsymbol{\psi}}_{C_1}^T & \cdots & \hat{\mathbf{p}}_{C_n}^T & \hat{\mathbf{v}}_{C_n}^T & \hat{\boldsymbol{\psi}}_{C_n}^T \end{bmatrix}^T \in \mathbb{R}^{(15+9n)\times 1} \quad \cdots(5)$$

- Time propagation
  - Linearized continuous time model for the IMU error state :

$$\dot{\hat{\mathbf{x}}}_{IMU} = \mathbf{F}\hat{\mathbf{x}}_{IMU} + \mathbf{G}\mathbf{n}_{IMU} \quad \cdots(6)$$

$$\mathbf{F} = \begin{bmatrix} \mathbf{0}_{3\times3} & \mathbf{I}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} \\ \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & -[\boldsymbol{f}_n]_\times & -\hat{\mathbf{C}}_b^n & \mathbf{0}_{3\times3} \\ \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & -\hat{\mathbf{C}}_b^n \\ \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{b}_a \cdot \mathbf{I}_{3\times3} & \mathbf{0}_{3\times3} \\ \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{b}_g \cdot \mathbf{I}_{3\times3} \end{bmatrix}, \mathbf{G} = \begin{bmatrix} \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} \\ -\hat{\mathbf{C}}_{b,k}^n & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} \\ \mathbf{0}_{3\times3} & -\hat{\mathbf{C}}_{b,k}^n & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} \\ \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{I}_{3\times3} & \mathbf{0}_{3\times3} \\ \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathbf{I}_{3\times3} \end{bmatrix}, \mathbf{n}_{IMU} = \begin{bmatrix} \mathbf{n}_a^T & \boldsymbol{n}_g^T & \boldsymbol{n}_{b_a}^T & \boldsymbol{n}_{b_g}^T \end{bmatrix}^T$$

  - P covariance matrix

$$\mathbf{P}_k^- = \begin{bmatrix} \boldsymbol{\Phi}_{k-1}\mathbf{P}_{II_{k-1}}^+ \boldsymbol{\Phi}_{k-1}^T + \mathbf{Q}_{k-1} & \boldsymbol{\Phi}_{k-1}\mathbf{P}_{IC_{k-1}}^+ \\ \mathbf{P}_{IC_{k-1}}^{+T}\boldsymbol{\Phi}_{k-1}^T & \mathbf{P}_{CC_{k-1}}^+ \end{bmatrix} \quad \cdots(7)$$

where $\boldsymbol{\Phi}_{k-1}$ is discretized matrix of $\mathbf{F}_{k-1}$

MULTI STATE CONSTRAINT KALMAN FILTER

- Measurement Model
  - Residual are typically defined as follows : $\mathbf{r} = \mathbf{H}\hat{\mathbf{x}} + \text{noise}$ ⋯(8)
  - As with the defined MSCKF's state vector, measurements are also determined by the camera pose with respect to the camera frame.

$$\mathbf{z}_k^{(j)} = \frac{1}{Z_j^{C_k}} \begin{bmatrix} X_j^{C_k} \\ Y_j^{C_k} \end{bmatrix} + \mathbf{n}_k^{(j)} \quad \cdots(9)$$

The feature position with respect to the camera frame, $\hat{\mathbf{p}}_{f_j}^{C_k}$ is given by :

$$\mathbf{p}_{f_j}^{C_k} = \begin{bmatrix} X_j^{C_k} \\ Y_j^{C_k} \\ Z_j^{C_k} \end{bmatrix} = \hat{\mathbf{C}}_b^c \hat{\mathbf{C}}_n^b (\hat{\mathbf{p}}_{f_j}^n - \hat{\mathbf{p}}_{C_k}^n) \quad \cdots(10)$$

$$\begin{aligned} \mathbf{r}_k^{(j)} &= \mathbf{z}_k^{(j)} - \hat{\mathbf{z}}_k^{(j)} \\ &\cong \mathbf{H}_{\mathbf{x}_k}^{(j)} \hat{\mathbf{x}} + \mathbf{H}_{f_k}^{(j)} \hat{\mathbf{p}}_{f_j}^c + \mathbf{n}_k^{(j)} \end{aligned} \quad \cdots(11)$$

However, error of $\hat{\mathbf{x}}$ and $\hat{\mathbf{p}}_{f_j}^c$ have the correlation, thus Eq. (8) is not in the form.

To apply for measurement updates, residual $\mathbf{r}_o^{(j)}$, by projecting $\mathbf{r}^{(j)}$ on the left nullspace of the matrix $\mathbf{H}_f^{(j)}$ is defined as Eq. (12).

$$\begin{aligned} \mathbf{r}_o^{(j)} &= \mathbf{A}^T \left(\mathbf{z}_k^{(j)} - \hat{\mathbf{z}}_k^{(j)} \right) \cong \mathbf{A}^T \mathbf{H}_{\mathbf{x}}^{(j)} \hat{\mathbf{x}} + \mathbf{A}^T \mathbf{n}^{(j)} \\ &= \mathbf{H}_{\mathbf{o}}^{(j)} \hat{\mathbf{x}}^{(\mathbf{j})} + \mathbf{n}_o^{(j)} \end{aligned} \quad \cdots(12)$$

MULTI STATE CONSTRAINT KALMAN FILTER

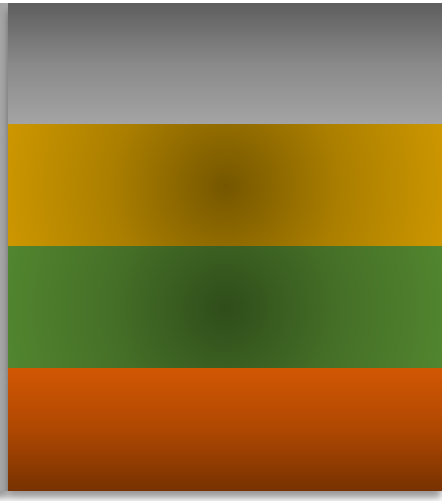
- Measurement Update
  - When a feature that has been tracked in a number of images is no longer detected, then all the measurements of this feature are processed.
  - Every time a new image is recorded, a copy of the current camera pose estimate is included in the state vector.

$$\mathbf{K}_k = \mathbf{P}_k^- \mathbf{H}^T (\mathbf{H}^T \mathbf{P}_k^- + \mathbf{R})^{-1} \quad \cdots(13)$$

$$\hat{\mathbf{x}}_k^+ = \mathbf{K}_k \mathbf{r}_k \quad \cdots(14)$$

$$\mathbf{P}_k^+ = (\mathbf{I}_{15+9n} - \mathbf{K}_k \mathbf{H}) \mathbf{P}_k^- (\mathbf{I}_{15+9n} - \mathbf{K}_k \mathbf{H})^T + \mathbf{K}_k \mathbf{R} \mathbf{K}_k^T \quad \cdots(15)$$

Future Work



Future Work

- Checking Feasibility
  - Conduct simulations based on datasets from high-altitude flights through open source.
  - The algorithms worth considering for verification are VINS-Mono and MSCKF-Mono.
- Take the experiment datasets with flying drone.
  - 50m, Constant altitude
  - 100m, Constant altitude
  - Maximum height 150m, varied altitude
- Performance analysis using the proposed techniques for each scenario.

Q & A

